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Artifacts in the co-production of knowledge in social design

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Abstract: Social Design Research is a design and research approach that is characterised by an attempt to make a difference for marginalized, underprivileged or vulnerable groups in society through participatory processes and the use of knowledge artifacts. This article demonstrates how knowledge is generated co-productively and shows the interplay between different types of tacit or explicit forms of knowledge and models the knowledge exchange between design researchers, practitioners and the participating citizens.

Keywords: social design; practice-based research; research through design methodology

1. Introduction

Accounting for knowledge claims in practice-based social design is essentially relying on addressing two basic research questions: *How do processes of making and designing of artifacts serve as a legitimate method of inquiry?* As participatory approaches are integral to these processes in social design (Armstrong et al., 2014), the second question is: *What kind of knowledge is co-produced when designers researchers make research artifacts in collaboration with practitioners and people we design with and for?*

We start out by going through existing accounts of how the making of artifacts can be used as a method of inquiry in practice-based design research. This related work section leads to the observation that artifacts are most often looked upon narrowly from the perspective of the design researcher while eschewing the perspectives of others participating in the research. Moreover, typically the kind of knowledge appreciated as the outcome of using artifacts in design research is limited to discussing different artifact-theory nexuses (e.g. artifacts spurring the development of conceptual frameworks, guidelines, implications for design, artifacts as exemplars sensitising philosophical concepts, etc.). Yet, as we shall argue, knowledge may take a variety of forms – theory being just one of them. Knowledge in social design is not something to be extracted from the artifact, but insights that emerge through an ongoing and nonlinear infra-structuring of situated practices and exchanges among research participants (Björgvinsson et al., 2012). To fathom this, it can be helpful to "can open"

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the ultimate particularity of case projects to lurk into how artifacts, practices, and knowledge are deeply entangled. In this article we delve further into two case projects. The first case project concerns a design game developed for children of prisoners in collaboration with family consultants working within the Danish Prison and Probation Service's parenting programme. The second case concerns a design probe developed for cancer patients in collaboration with nurses and health professionals. Both cases serve as exemplars to demonstrate how research artifacts in social design projects play a central role as they allow for different kinds of knowledge co-production in communities of practices.

2. Knowledge production in social design research

In design research much attention has been dedicated to describing how artifacts may serve as vehicles for knowledge production. Referring to Carroll and Kellogg's (1989) foundational conception of artifacts as theory nexus, Haynes and Carroll (2007) argue that artifacts can be considered as embodying hypotheses that can be tested and lead to new knowledge. In addition to this first role of the artifact, Wensveen and Matthews (2014) identify three more roles that artifacts can play: Artifacts can be designed as *research prototypes* that are used to generate understanding of use contexts and users (e.g. probes and design games). Artifacts can also be deployed as *research archetypes* that increase knowledge of new concepts and design spaces, e.g. Wensveen's alarm clock designed as an archetype of emotional design (see Overbeeke & Wensveen, 2003); and the fourth role is when prototyping of artifacts become a *vehicle of inquiry*, where questions about how the artifact itself was made and crafted is foregrounded, answered and written on. To this typology, a fifth role can be added – *artifacts as discourse* (Seago & Dunne, 1999) – as when artifacts are designed to encourage critical reflection and debate (e.g., Dunne & Raby's Faraday Chair as a discursive design object prompting reflection on the effect that magnetic fields and electrical currents may have on people's everyday life).

Accounts like these go hand in hand with accounts that strive to unravel what kind of knowledge is gained from designing artifacts. In the attempt to show that practice-based design research fully lives up to the criteria of being a scientific discipline, theory has often been regarded as the most privileged form of knowledge outcome. For instance, Zimmerman et al. (2010) demonstrate how theory can result from research through design in the form of conceptual frameworks, guidelines, implications for design, etc. Accounts like these, however, have spurred controversy, leading Gaver (2012) and Bowers (2012) to warn against scientising the approach. In their view, theory underspecifies the ultimate particularity of design work, which should be appreciated as the primary outcome. If theory has a value for design research it is in the form of annotations - aesthetic principles or criteria derived from the designer's portfolio. Recently, attempts have been made to even out the discrepancy between these positions by showing how the making of artifacts gives rise to particular forms of theory building deeply rooted in practice (Markussen, 2017; see also Redström, 2017).

While these accounts appear to fully answer the first basic research question that was introduced above, they are not sufficient to understand the knowledge claims in social design. We need to keep in mind that theory is only one form that knowledge can take (although an important one in research). Moreover, the accounts examined so far are 'narcissistic' in the sense that they deal with knowledge as if it emanates exclusively from design practice and has value only for the researcher(s). Typically, in social design, design practice involves several other practices and the collaboration with marginalized or vulnerable groups, meaning that the two research questions cannot be answered independently of each other. Artifacts are thus used in the co-production of knowledge, the form of which may vary depending on how different practices and needs are orchestrated and negotiated.

If knowledge is co-produced, the notion that it is somehow extracted from design practice by the researcher must be expanded. Rather, knowledge is brought about through knowledge exchanges among design researchers, practitioners, active citizen groups, organisations, decision-makers, and others who engage in a community of practices (Sangiorgi, 2011). In design research, the notion of ´boundary objects´ has been borrowed from STS-studies (Star, 1988) precisely to show how objects may allow for interpretive flexibility among members of a community of practices who represent different social worlds and who act toward and with the object (Star, 2010, s. 603).

Yet, what is important to notice in social design is that the collaborative acting with the object is bound up with the idea of acting socially responsive to target underserved groups (Thorpe & Gamman, 2011). Hence, as Campbell and Vanderhoven (2016) have convincingly argued, knowledge claims in social design are tightly coupled with the need for assessing the relevance of acting and its societal impact. This is achieved not only through design outcomes, but also through the nonlinear participatory design processes leading up to the outcome. Elsewhere, we have demonstrated how relevance and impact resulting from such processes can be conceived of in terms of 'ripple effects' that benefit people, organisations and design researchers (Knutz & Markussen, 2019).

In this paper we shall focus in more detail on knowledge. More specifically, we suggest making a distinction between different forms of knowledge that can be generated co-productively in social design projects. *Practical knowledge* may be as important as theoretical knowledge, as it enables designers and practitioners to build capacity in order to prompt actions towards social change. *Experiential knowledge* is another kind of knowledge that may help vulnerable or marginalized people to better understand and cope with a difficult life situation. *Social knowledge* in relation to the individual or group is the third type of knowledge form we have identified as relevant for the social design case projects that we present in this paper.

3. Forms of knowledge: Practical, experiential and social

One fundamental distinction of knowledge is *explicit* knowledge as opposed to *tacit* knowledge. Tacit knowledge is challenging since it cannot be explicitly represented and often remain inarticulate (Chua, 2002; Kakabadse et al., 2003; Polanyi, 1983). It covers a broad range of meanings, but often used examples are skills, emotions or experiences that people have gained through personal or professional experiences, for instance, of riding a bicycle, interacting with people, using tools and materials in a particular way or explaining the use of language. Basically, it contains everything a person knows how to do, but may not be able to explain.

Important for the production of knowledge are the separate ways in which we acquire knowledge and how that relates to the knowledge outcome. In this article, we take as our starting point knowledge that is co-created and requires an interplay between different types of tacit or explicit knowledge forms. Most notably, we distinguish between *practical knowledge, experiential knowledge and social knowledge.*

Practical knowledge is knowledge acquired through daily activities or professional practices, and it is acquired by striving to perform a task; the knowledge outcome is about *knowing how* (Heron, 1992, 1996). While *practical Knowle dge* is gained by doing things, individually or as a team, theoretical knowledge is gained by studying, for example, theoretical models, texts, design manuals, computer codes or musical notes. Although theoretical knowledge may enable you to understand the basic concepts and have knowledge of how something should work theoretically, it cannot be transferred seamlessly to practice without exercises and real-world practical experiences.

Experiential knowledge outcomes are gained through transformations of being and experience (Heron, 1996: 41). In health sciences, for instance, it has been recognized that patients' experiential knowledge can contribute to an improved quality and relevance of health research. Especially patients' experiential knowledge and embodied lay-expertise are obtained as a result of changes in their experiences of the disease, the treatment they receive and the physical and mental changes they go through (Blume, 2017; Caron-Flinterman et al., 2005). This experiential knowledge can supplement theoretical knowledge by providing new insights and perspectives (Entwistle et al., 1998; Goodare & Lockwood, 1999; Popay & Williams, 1996).

Social Knowledge concerns knowledge of about social groups held by an individual or by the group itself (Linde 2001: 161). It may include knowledge about what it means to be a member of a particular group and the identity of the group as well as individual knowledge about one's identity and personal history (Camic et al., 2011; Linde, 2001). McLean (2017) argues that identity can be conceived as a relational phenomenon and that it is co-constructed through a process involving stories we tell about ourselves or others or stories that others tell about us. Linde (2001) argues that narratives and storytelling can be central means by which social knowledge is conveyed and that narratives can bridge the gap between the tacit

and the explicit, allowing tacit social knowledge to be expressed (told, visualized, performed) and learned without necessarily being articulated in propositional form (*knowing that*).

In the following we wish to demonstrate how different forms of knowledge can be generated co-productively in social design projects. The first research artifact we will use as an exemplar is a design-game called the 'the story-card game'. Knowledge in this project is engendered through knowledge exchanges between design researchers, family consultants and children of prisoners.

The second research artifact we will use is a probe referred to as 'the body probe'. In this project the knowledge exchanges take place between design researchers, patients and health professionals, including nurses and health instructors working with care practices and stress relieving activities.

4. Case 1: A design game for children of prisoners

4.1 Research context

The empirical material that we draw upon in case 1 has been collected as part of the social design project *Changing Perspectives* (2017-19), which aims at developing a playful tool for dialogue used by family consultants and prisoner fathers within the Danish Prison and Probation Service's Parenting Program. Table 1 provides an overview of the main design research activities in the project.

For the purpose of this article, we are using the data collected in research activity 2, which concerns a design game workshop held at a centre for family counselling. The purpose of this workshop was to understand children's experiences and challenges related to having a father in prison and to promote the sharing of stories through visual storytelling and the use of design games. Subsequently, the children's stories were anomymised, altered and brought into family consultants' conversations with prisoner fathers in order to falitate nuanced discussion of parenting roles.

Applying design games as a design ethnographic method is particularly useful when involving children in participatory research activities – as it makes use of play, game rules, fiction and narrative elements - and through this, it allows children to play out and rehearse various scenarios that relates to their past, their everyday life or potential or imaginative futures (Dindler, 2010; Knutz et al., 2016).

The *story-card game* was designed as an ethnographic tool to gain insights into children's challenges of having a father in prison as well as to assist them in the process of articulating and sharing experiences through personal and collective storytelling in a different way (see gameplay below). The sharing of stories plays a central part in the Danish Prison and Probation Service's Parenting Program. Some of the program's family consultants are trained in systemic family and narrative therapy and they refer to particular narratives as *core stories* or *universal stories*. These are stories told by prisoners that manifest recurrent issues and

themes, which are recognisable for many prisoners, their families and family consultants (e.g. stories of loss, loneliness, separation, guilt, secrecy, etc.). By working with these core stories, it is possible for the family consultants to separate the individual from the problem and to work with common challenges through therapeutic group meetings in the parenting program. However, most core stories reflect prisoners and not their children's perspectives. Hence, one of the aims of the *Changing Perspectives* project was to collect core stories told by children and explore how they could pontentially be integrated into family consulty practices in the parenting program. For this purpose several research activities was carried out, among them the game workshop.

Table 1. Outline of the research activities of the entire research project. In this study we use the empirical data from activity ; The "storycard game" workshop, involving 8 children of prisoners

		Method	Aim of Study	Type of Data	Participants	Activities	Knowledge Outcome	
Explorative Workshops	Doll-Scenario workshop	Design Ethnography & Co-design	To understand the therapeutic experience of working with father groups in prison	Empirical materials from design games Video recordings	8 family councellors 2 researchers	Sharing theraphy experiences through the use of doll scenario games	Understanding practices of father-groups in prisons. Understanding father's dilemma from a therapeutic perspective	Actvitity 1
	Story card workshop	Design Ethnography & Co-design	To understand children's perspectives of having a father in prison	Empirical materials from design games (children's drawings)	8 children of prisoners 2 family councellors 2 researchers	Sharing stories of having a father in prison through design games & visual storytelling	Understanding children's dilemma more profoundly through the "thickening" of stories"	Actvitity 2
	Roleplay workshop	Design Ethnography & Co-design	To evaluate children's stories. To co-design different type of game mechanics.	Video recordings of roleplay games Fieldnotes	4 family councellors 3 researchers	Indentifying core-stories. Developing game play & gamerules through roleplay	Understanding different perspectives on children's stories from a therapeutic perspective	Actvitity 3
Pilottesting	Group interviews	Focus groups Interview	To understand therapeutic practice without game	2 recorded and transcribed group interviews	6 family councellors 3 researchers	Sessions of focus group interviews with family councellors	More profound understanding of therapeutic practice	Actvitity 4
	Rehearsing workshop	Participatory Observation	To understand therapeutic practice with the new game	Video recorded game sessions and talks	8 family councellors 4 researchers	Roleplay games to practice and evaluate game prototype before intervention	insight into the different therapeutic practices when activating game mechanics.	Actvitity 5
Intervention ->		*8 family councellors and fathers' groups make use of the game prototype in prison*						
Evaluation study	Self-reporting through audio	Self-report	To evaluatue the experienced intervention	4 audio recorded self-reports	4 family councellors	Conducting audio recorded self-reports immediately after intervention	Insight into to how the family councellors experienced the intervention in prison	Actvitity 6
	Group Interviews	Focus group Interview	To evaluate game prototype with imprisoned fathers	4 recorded and transcribed group interviews	7 imprisoned fathers 1 researcher	Group interviews in prison with fathers who have played the game	Knowledge about father's perspective on parenting roles and the game's potential in supporting this	Actvitity 7
	Collecting anonymous feedback	Feedback through Questionnaire	To evaluatue game prototype with family councellors	8 completed questionnaires	8 family councellors 2 researchers	Collection of anonymous feedback through questionnaires	Insight into how the family councellors evaluted the game's potential in supporting parenting roles	Actvitity 8

4.2 Format of game workshop

Eight children of prisoners (age 9 -14 years) participated in the game workshop, together with two family consultants, one assistant and two design researchers. All the children were boys. The workshop lasted half a day and was part of a weekend program that consisted of several types of therapeutic and social activities such as therapeutic group sessions, walks, football, table tennis, cooking and eating together. The format of the game workshop was planned together with the family consultants, as it needed to fit into the overall (social and therapeutic) program.

4.3 Materiality and gameplay of the story-card game

The narratives in *the story-card game* are informed by previous workshops held with children and family consultants and all use the same starting point: focusing on what it's like to have a father in prison seen from a child's perspective. All cards had the same cartoon-like form, using four frames each with a drawing and a text field to convey a coherent narrative (see Fig. 1).



Figure 1. Story-cards and blank cards with the same structure of 4 frames and a text field.

Apart from picture stories the design game contained a pile of empty cards with the same structure of 4 frames and a text field as well as drawing equipment for each of the children. The 'task' for the children was to create stories/game content by either reshaping the existing stories or draw new picture stories that they felt were missing. In doing so, these stories 'thicken', meaning that the same narratives are being reflected on, changed, broadened, adjusted or added to with new stories that are all circling around the same theme: growing up with a father in prison.

After having created their own game cards (using the empty cards + drawing materials) the game could start: All cards are mixed, each player gets 7 cards, the rest are spread out face down on the table. The task for the players is to collect stories that belong together (= 4 cards) and to get rid of all 7 cards. If a story is collected, the player puts the 4 cards down on the table and "show" and "tell" the story. It need not be the same player who shows and tells: the player who puts down the story can tell the story or the player who actually illustrated the story can tell the story. If a story is too difficult to tell for the person who drew

the story, the family consultant steps in and facilitates the telling of the story based on the illustrated playing cards.

4.4 One example

One boy, Ben, drew a story entitled “Bad Christmas”. It was about an event that took place shortly before Christmas and which resulted in his mum being hospitalized, his father being imprisoned and he and his siblings having to spend Christmas with his aunt and uncle. Ben clearly claimed authorship of the the story, when it was collected by a player in the game session, but he wanted one of the family consultants to verbalised and retell it. After the session the family consultants explained to the design researchers that it was the first time that Ben had been able to express elements from his personal story and share it with the rest of the group, although not verbally articulating it himself. The family consultant pointed to the fact that in Ben’s story, his mother’s hospitalisation was drawn as one incident happening after his father’s arrest. What was too difficult for Ben to explain - and what the family consultant was careful not to disclose in the game session - was a traumatic causal, narrative link between the two events: His father was arrested for attempted murder on his mother, leaving her severely wounded from a knife attack. In this case, the boy’s personal story was visualised and brought up in conversation through the game session, while part of the family consultant’s background knowledge was deliberately not voiced. This is an example of practice research where design of artifacts play a vital role in the co-production of various forms of knowledge.

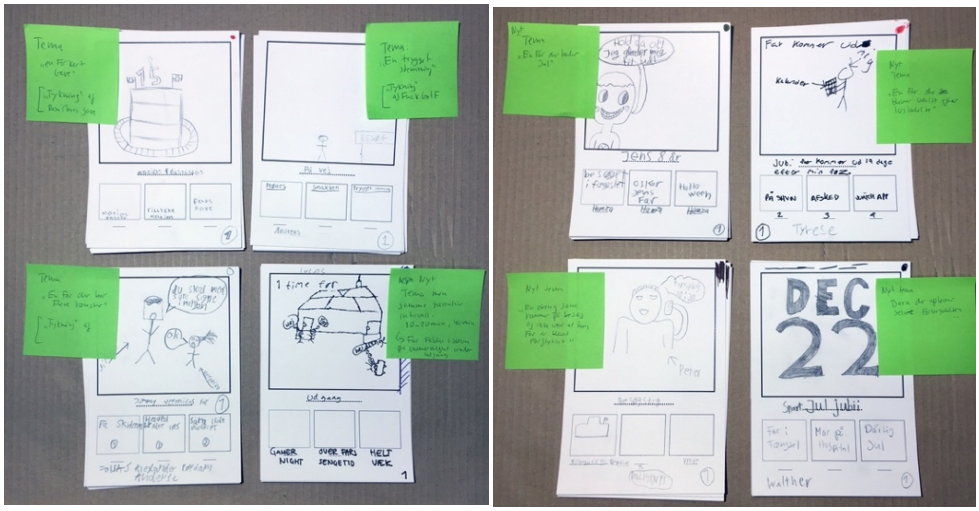


Figure 2. The eight children of prisoners (age 9 -14 years) produced each their set of cards

4.5 Co-production of knowledge

The material form of the design game is pivotal as the creation of picture stories allows tacit knowledge of prisoners’ children to become explicit (as in Ben’s case). But the picture stories can also serve the purpose of children reshaping existing stories in the game made by other

children thereby providing multiple new perspectives on stories, that is the thickening of existing stories (for instance about deprivation, new family patterns, rituals around holidays, challenges during visits, powerlessness, anger, violence and crime). By making use of different tactics of "making" stories, the boys offered different ways to make explicit and share social knowledge about their belonging to a group, that has one thing in common: they are all members of prisoner families and have to learn how to cope with the collateral effects of crime and paternal incarceration.

The design game is informed by and co-designed with family consultants who accepted integrating the game workshop to support children, in talking about what it means to be a member of a prisoner family. In carefully guiding the children in how to share their social knowledge, the consultants draw upon their practical knowledge. This is not an easy task. The picture story “Bad Christmas” vividly illustrates that some children’s narratives can be brutal and difficult for children to articulate. In such situations the therapeutic expertise and know-how of the family consultants is an important part of the process of scaffolding dialogue. But Ben’s incidence also shows, how the use of the design game may call for a broadening of the consultants’ practical knowledge. Notably, as the game seemed to have made Ben comfortable for the first time in sharing his story, it invites consultants to consider changes in therapeutic practice. A change perhaps, from inviting children to tell a story, to draw a story, using means that children are familiar with (sketching, colouring, cartoon-language etc.).

The design research team made use of their practical knowledge in terms of designing a game that needed to align with children’s storytelling capabilities and the therapeutic practices of the family consultants. Moreover, the use of the design game fosters a mutual knowledge exchange in the sense that the game workshop call for carefully considering how research artifacts used in social design practices may position vulnerable participants into certain roles and situations that require professional handling and critical awareness. Ben’s sharing of his story required utmost care and sensitivity on behalf of the consultant. The evaluation that it was good for him to share it in the group, also relies on the belief that children expressing their story is to the benefit of their well-being, which may not always be the case. Issues like these point towards the design researcher developing ethical implications as an integral part of their practical knowledge.

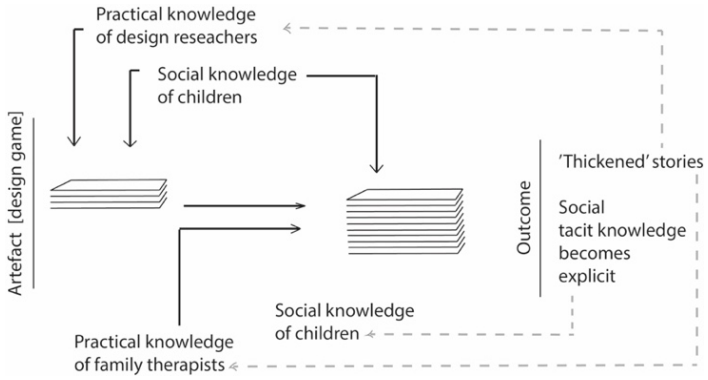


Figure 3. Knowledge exchange in relation to the story card game

In summary, Figure 3 depicts the knowledge exchanges between researchers, family consultants and children. The design researchers apply their design and research skills to build a design game that can collect stories of children; the children participate by visualising identity stories or new perspectives by drawing on their social knowledge. In performing and practicing the design game the family consultants extends their practical knowledge of how to work with children of prisoners. The knowledge outcome points in three directions: it adds to the practical knowledge of the therapeutic practice; it also provides practical knowledge for the design research team in working out ethical implication of the method as well as gaining insight into children's perspectives and finally it allows tacit social knowledge to surface. The latter might influence the child's individual social knowledge.

Three aspects are important in the process of knowledge exchange. First, the shift in practice, from telling a story to drawing a story, using means that children are familiar with (drawing, sketching, colouring, cartoon-language etc.). Secondly, the perspective-taking part, which means that the children could either choose another child's story and tell/draw it from their own perspective, or they could create a new story. Third, the sharing of stories, which makes it possible for the children to reflect on their own as well as other children's stories.

5. Case 2: Tracking embodied emotions through probing

5.1 Research context

The second research artifact and related study we will use as a case example, is part of an EU-funded research project focusing on Patient Empowerment (2015-2021). The aim of this project is to co-design and develop methods and concepts for cancer patients to support them in navigating and aligning selfcare and self-management with medical treatment. The project involves years of field studies, as well as workshops and probe studies with health professionals and cancer patients (Knutz et al., 2014, 2017; Knutz & Markussen, 2019). Table 2 provides a short-listet overview of the design research activities in the project.

For the purpose of this article, we are using the data collected in research activity 2, which concerns a probe study at a health centre for cancer patients. The aim of this study is to obtain insight into how to convey or express the difficult feelings and emotions people go through when they are diagnosed with a life-threatening disease. We refer to this probe study as the "body probe".

Using probing as design approach - as well as an ethnographic method - is well described in design research literature (Boehner et al., 2014; Crabtree et al., 2003; W. W. Gaver et al., 2004; Hemmings et al., 2002; Mattelmäki, 2006, 2008; Mattelmäki et al., 2016; Wallace et al., 2013). Mattelmäki (2006; 2008) identifies four purposes for using probes and refer to these as: (1) inspiration for enriching the design process; (2) collecting information; (3) user participation in ideation; and (4) dialogue to build up understanding and interaction between users and designers. In this study, probing is used as a collaborative inquiry which means that the research team has participated with the patients in the probe sessions. This particular approach to probing has been applied in other activities in the project (see activity 3) and has been described in previous articles (Knutz et al., 2018).

Table 2. Outline of the research activities of the entire research project. In this study we use the empirical data from activity 2, the "Body Probe" study involving 20 cancer patients.

		Method	Aim of Study	Type of Data	Participants	Activities	Knowledge Outcome	
Explorative Fieldwork	Fieldwork at Hospital	Observation & Roleplay	To understand how decision-making is practiced by medical staff	Fieldnotes Photos Video recordings	6 patients 1 doctor 1 nurse 2 researchers	Recordings of 6 doctor-patient consultations . Roleplay activities with staff	Insights into how shared decision-making is practiced in hospitals as well as politics of participation	Actvitity 1
	Probe studies	Probing & Participatory Observation	To understand how cancer feels. To explore new methods that fits into the patients healing lives	Probe-returns	20 patients 2 health instructors 2 nurses 1 researchers	Eksperimental collaborative probing combined with stress relieving activites	Insight into how cancer feels and how patients experience the subjective state of the body	Actvitity 2
	Probe studies	Probing & Participatory Observation	To understand good care. Probing needs and values of the patient	Probe-returns	8 patients recieving conventional, complementary and alternative treatment	2 probe studies based on collaborative probing as well as probing based on self-report	Insights into patient perspectives, self-management and what works for and against good treatment.	Actvitity 3
Co-designing	Co-design workshops	Participatory workshops & Visual data mapping	To co-design tools and tactis of selfcare To align selfcare with medical care	Emperical material from participatory workshops	10 patients 10 researchers 10 health professionals 2 researchers	3 workshop sessions 4 data mapping sessions	Insights into selfcare. A co-designed prototype that aligns selfcare with medical treatment	Actvitity 4
	Evaluative workshop	Evaluative workshops	To evaluate patients tactis of selfcare and strategies for implemention	Emperical material from participatory workshops	20 health professionals and patients from Denmark & Germany	2 workshop sessions	Insights into selfcare. A re-designed prototype: a kit that contains selfcare strategies	Actvitity 5
Intervention ->		*10 patients and 8 health professionals lives with/explore a prototype that that contains 15 tools, tactics and selfcare strategies, for cancer patients*						
Evaluation study	Self-report & video interviews	Video recorded self-report & Interview	To evaluate prototype with patients	5 recorded assesments	5 patients 2 researchers 3 researchers	Conducting video recorded assesments after intervention	Knowledge about patients perspectives on selfcare	Actvitity 6
	Self-report & video interviews	Video recorded self-report & Interview	To evaluate prototype with patients & health professionals	5 recorded assesments	5 patients 1 doctor 1 nurse 2 researchers	Conducting video recorded assesments after intervention	Knowledge about patients and health professionals perspectives on selfcare	Actvitity 7
	Interview & Questionair	Feedback through Questionnaire	To evaluate prototype with health professionals	Interview data & questionnaires	1 doctor 1 nurse 3 psycologists 1 health worker 1 researcher	Online group interview combined with questionnaire	Knowledge about patients and health professionals perspectives on selfcare	Actvitity 8

The *body probe* is designed to assist the process of self-report, under difficult circumstances (during cancer treatment). Nurses working with health communication and health research often use data collection techniques that are based on interviews, questionnaires or variations thereof. The challenge with these methods is, that if patients are not physically or emotionally able to put their feelings into words then these approaches have limited use. To overcome this barrier, nurses and health professionals collaborated with design researchers, to develop a different approach. The idea was to combine a probe study with a stress relieving activity and develop a new method that would fit into the social and healing life of patients.

5.2 Format of probe study

The *probe study* was thus embedded in a stress relieving activity - a therapeutic course on dynamic movement practice referred to as '5Rhythms', originally developed by Gabrielle Roth in the late 1970s. This practice consists of training patients in particular body movements and physical exercises that make it more tolerable to accommodate emotions associated with, for instance, grief, pain, anxiety or serious side effects of chemotherapy. Twenty patients participated together with two instructors and one design researcher. The latter participated in all exercises and received training at the same level as the patients. The therapeutic course as well as the probe study lasted 9 weeks (2 hours per week) and took place at the Centre for Cancer Patients, next to a large danish hospital.

5.3 Materiality and interaction qualities of the body probe

Central to the dynamic practice of movement is the concept of flow, which refers to the notion that everything is energy and moves in waves, patterns and rhythms. The material content of the *body probe* was aligned with this concept by adjusting the movement qualities (associated with the different rhythms) with the interaction and tangible qualities of the probe material. These were organised in a box (see Figure 4).

In one compartment there were rigid materials that could be broken or stacked (wooden sticks, oblong pieces of cardboard); in the second there were loose materials that could be stretched, tied or twisted (elastic, string, thread, reed); the third contained malleable materials that could be modelled or bent (clay, aluminium, steel wire) and in the last compartment there were soft materials that could be cut into smaller pieces or used as filling or as protection (cotton, wool, felt, foam, insulation fabric).

Furthermore, the kit contained a pen, a pair of scissor, tape and cardboard with 9 labels placed in separate pockets (Figure. 4, right).



Figure 4. The body probe consisted of a box with materials with different interaction qualities

The probe task for the participants was to use these materials as well as keywords to describe their bodily emotions immediately after each dynamic movement session. Thus, they were asked to self-track their emotions through 9 encounters. After 9 weeks, participants were invited to assess the effect the exercise had on their body and mind, through the use of a postcard.

The image below (figure 5) shows an annotated probe return. This photo is displayed to give an insight into how a patient's' self-tracking proceeds over time and how the state of the body is experienced and visualised through different uses of the materials.



Figure 5. Annotated probe return of a cancer patient

At the end of the 9-week course all patients had self-tracked an overview, indicating their emotional states during the course. These were documented and then personally handed over to all participants as an imprint of the process they had been through. Thirteen patients made use of the postcards. Out of these, seven patients concluded that the materials helped them in communicating their emotional states, three patients concluded that the materials had not helped them and three was not sure.

5.4 Co-production of knowledge

To self-track bodily experiences and emotional states, patients needed to draw upon their experiential knowledge. This knowledge is materialised through the use of keywords, abstract and figurative, forms, word images, drawings and symbols. This provides the research team with evidence of what a person affected by cancer goes through when attempting to access his/her personal emotional life and engage in expressing and documenting it in a self-report. This knowledge is valuable from a research perspective to qualify the method, and it can also inform the further use of probes in nursing and care practices.

To guide patients through the dynamic movement practices - and thus engage and train patients in how to relate and listen to their bodily emotions through movement ('5Rhythms'), the health instructors make use of their practical knowledge.

The design research team made use of their practical knowledge in the design of a probe, that is informed by the concept of "dynamic movements". As materialised artifact, the probe plays a central role in expressing felt emotions and allowing tacit experiential knowledge to become explicit.

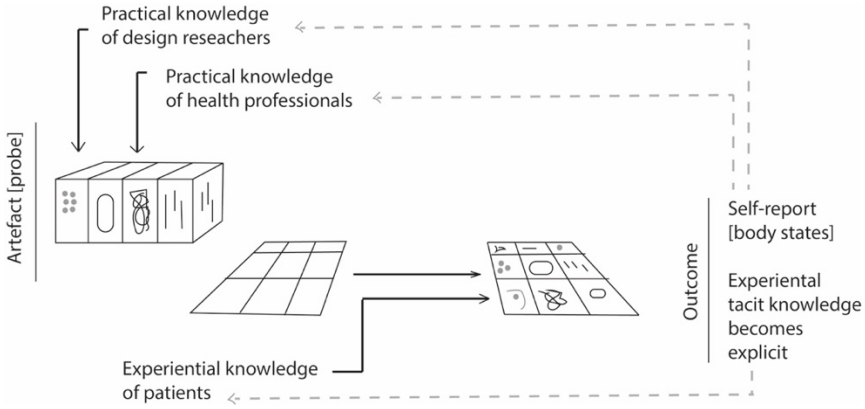


Figure 6. Knowledge exchange in relation to the body probe

Figure 6 depicts the knowledge exchange between design researchers, health practitioners and patients. The design researchers apply their design and research skills to create a probe kit that can self-track bodily emotions; the patients participate by 'probing' their embodied experiences – by drawing on their experiential knowledge. This process is guided by the health practitioners, who participate with their practical knowledge of working with patients. The knowledge outcome points in three directions: it adds to the practical knowledge of nursing and caring practices; it provides practical knowledge for the design research team in qualifying the method of probing in a highly experimental set-up and finally it allows tacit experiential knowledge to become explicit and adds to the embodied expertise of patients.

In this process of knowledge exchange, an *exchange of practice* is also taking place: First, the shift in practice from expressing emotions through movements and rhythms (what goes on inside the body, internalizing) to the expression of emotional states and thoughts through the use of probe materials (to communicate and make explicit). Secondly, the serial self-

tracking part, which enables the patients to track or express their emotion over time. These reflections describe states of the subjective body during a temporal process. As knowledge exchange (in the form of emotional mappings) it is adding to the embodied expertise that patients gain, as a result of their experiences with cancer, and the changes they feel is happening.

6. Concluding remarks

In a design research process, artifacts can take up several roles along the process of exploring, designing, intervening and implementing. The research artifact (as material form) may stay the same, but it can be used in different forms of explorations and be made subject to different forms of analysis and reasoning (data-driven, hypothesis-driven, situation-driven), which may eventually lead to different types of knowledge.

In case 1 the design game (story card game) take-up the role as a design ethnographic tool for the design research team and contribute with knowledge about the children of prisoners and the context designed for. For the family consultants and the children, the artifact assists the therapeutic process of sharing and scaffolding stories through visual storytelling.

In case 2 – the body probe acts as an embodied hypothesis – testing new data collection technique for nurses to be applied in health communication and for the design research team as a highly experimental and situated probe technique. For the health professionals and the patients the artifact becomes part of stress relieving activity that may help patients to communicate their emotional states

In this paper we have explored the role of artifacts in relation to process of knowledge exchange among different participants (the design researcher being one of them) and we have exemplified how this knowledge can be understood as an interplay between practical, experiential and social knowledge. The two models presented (figure 3 and 6) are broad examples of how to diagram such an interplay.

In doing so, we contribute to the typology of Weensveen & Matthews by expanding the role of artifacts to include artifacts in socially co-producing knowledge that can benefit different communities of practices. This role is added to their framework in Table 3.

Through this table, we specify the use of artifacts in social design research projects, in which the collaborative effort and social responsiveness is central. Important for this type of research is that knowledge arises through an ongoing exchange of practices. And as demonstrated through the two case examples, this exchange may also lead to change of practice.

The material form and methodological application of the artifact play an important role, but cannot be limited to the artifact-theory nexuses. Other knowledge forms are at play, both in the process of prototyping and designing as well as in the process of applying and intervening. For instance, *practical knowledge-forms* may contribute with know-how in relation to *how* to design a particular design ethnographic tool whereas *social knowledge-forms* are crucial when, for instance, having to facilitate game sessions with children of prisoners or when

conducting collaborative probe-sessions with citizens in chemotherapy. Here the design researcher need to blend-in, adapt to the situation and become a member of that particular social group.

Table 3. Representing the expansion of Wensveen and Matthews'(2014) original typology

Role of artefacts	Research aim	Material Forms	Examples of knowledge outcome	References and case examples
Artefacts as *embodying hypotheses*	Testing and evaluating psychological claims and assumptions of user-interface interactions	…might be crafted as working or functional prototypes tried out in lab or field settings	Knowledge about conditions for improving usability and user experience	Carroll and Kellogg (1989) Haynes and Carroll (2007)
Artefacts as *research prototypes*	Generate ideas and understanding about use contexts and users	…might be crafted as cultural probes, design games and design ethnographic tools	Inspiration for design or information about the people and contexts designed for	Boehner et al (2014) Mattelmäki (2008)
Artefacts as *research archetypes*	Sensitize design practice to new concepts and theories	…might be crafted as exemplars of emotional design, ludic design, aesthetics of interaction, etc.	Knowledge about how new concepts may open up for unexplored design spaces	Overbeeke & Wensveen (2003)
Artefacts *as vehicles for inquiring* into the design practice	Exploring the process of designing, new materialities and technologies	…might be crafted as artistically inclined experiments and design artefacts	Implications for design or guidelines	Persson (2013)
Artefacts as *discourse*	Encourage critical reflection and debate	…might be crafted critical design or adversarial design	Knowledge about political agendas or hidden ideologies in design	Seago & Dunne (1999) Dunne (1999) DiSalvo (2012)
Artefacts used in *the co-production of knowledge*	Enable communities of practices to co-produce knowledge in order to prompt social change	might be crafted as social and participatory design activities or artefacts	Knowledge about *exchange* among design researchers, practitioners and citizen groups in society	

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